MIL-F-47222(MI)
12 July 1974
SUPERSEDING
MIS 16282A
4 November 1968

MILITARY SPECIFICATION

FOAM, POLYURETHANE, RIGID

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of gas-expanded, isocyanate-hydroxl reaction formed rigid polyurethane foam.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-810	Environmental Test Methods

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

FSC 9330

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American Society for Testing and Materials

ASTM C 273	Method of Shear Test in Flatwise Plane of Flat Sandwich Constructions or Sandwich Cores, Method of
ASTM D 1621	Compressive Strength of Rigid Cellular Plastics, Method of Test for
ASTM D 1622	Apparent Density of Rigid Cellular Plastics, Method of Test for
ASTM D 2127	Water Absorption of Rigid Cellular Plastics, Method of Test for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample of the polyurethane foam shall meet the requirements of this specification.

3.2 Data. Unless otherwise specified (see 6.2), the supplier shall provide detailed manufacturer's instructions for foam processing, including data on mold preparation, mixing and foaming procedures, time for self curing, mix ratio (formulation), and foam time with each lot of foam.

3.3 Material. Tha material shall be a rigid gas-expanded, isocyanate-hydroxl reaction product, closed-cell polyurethane foam and shall be the product of a two component, self-curing foam system. It shall be processed according to the manufacturer's instructions (see 3.2) using a foam machine capable of maintaining necessary control over temperature, ratio of components, and mixing.

3.4 Physical and mechanical properties. The physical and mechanical properties of the cured polyurethane foam shall be as specified in Table I.

3.5 Chemical properties. The polyurethane foam shall be the reaction product of polymethylene polyphenyl isocyanate and polyhydroxy polyether or a polyester material (preferably with aromatic or cyclic

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components as members of the chain structure). Appropriate cell control agents (surfactants) and reaction rate controllers (catalysts) shall be used as required to effect polymer buildup on molecule length crosslink density, viscosity, and cure. A blowing agent of carbon dioxide, generated by the reaction of the isocyanate group and water, shall be included in either of the two basic components to expand the polymerizing mass and impart a cellular (foam) structure to the cured polymers. The amount of blowing agent included shall be sufficient to form a cured foam conforming to the density requirements in Table I.

Table I. Physical and mechanical properties of cured polyurethane foam

Property	Values required
Density, pounds per cubic foot (pcf), free blown	12.0 ± 1.5
Density, pcf, pressure blown	22 ± 2
Compressive strength at 10 percent deflection, pounds per square inch (psi), minimum, at 22 ± 2 pcf density	900
Tensile strength at yield, psi, minimum, at 22 ± 2 pcf density	600
Shear strength, psi at plus 24 degrees centigrade (C) (plus 75 degrees Fahrenheit (F)), minimum, at 22 ± 2 pcf density	300
Water absorption, pounds per square foot of surface area per day, maximum, at 22 ± 2 pcf density	0.02
Humidity aging, dimensional changes, at 22 ± 2 pcf density	(see 4.5.8)
Environmental resistance (see 4.5.9), percent volume change, maximum, at 22 ± 2 pcf density	0.1 (No cracks or other visually perceptible deterioration)

3.6 Storage life. The shelf storage life of the foam system material in original containers shall be not less than 6 months in a sheltered area.

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3.7 Workmanship. The foam system components shall be uniform in quality and free from impurities and other defects that could adversely affect their use. The workmanship in the foamed items shall be commensurate with the requirements of the contract or purchase order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specificiation where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The examination and testing of the polyurethane foam shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.2.1 Preproduction tests. Preproduction tests shall be conducted only on the preproduction samples (see 4.3.1) and shall consist of all the examinations and tests specified herein except as specified in the procurement documents (see 6.2). Further production of the polyurethane foam by the supplier, prior to approval of the preproduction tests, shall be at the supplier's risk.

4.2.2 Quality conformance tests. The quality conformance samples of the inspection lot (see 4.3.2) shall be subjected to the tests of 4.5.1 through 4.5.8.

4.3 Sampling.

4.3.1 Preproduction sample. A preproduction sample (of sufficient quantity to perform the tests of 4.5) of the proposed polyurethane foam, manufactured using the same materials, methods, and procedures proposed for production quantity shall be prepared as specified in 4.4. Except for the sample used for free-rise density determination, the preproduction sample shall be foamed in a restricted, pressure-rise condition.

4.3.2 Quality conformance sampling. Unless otherwise specified (see 6.2), sufficient material shall be taken from each different lot (of foam system to be used by the foam supplier) to form samples for tests of 4.5.1 through 4.5.8. Except for the sample used for free-rise

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density determination, the material from each different lot shall be foamed in a restricted, pressure-rise condition. The inspection lot size shall be in accordance with MIL-STD-105 or other sampling plan as specified by the procuring activity.

4.4 Preparation of test specimens. A block of cured polyurethane foam shall be fabricated in accordance with the manufacturer's instructions using procedures proposed for production and shall be of sufficient size to provide specimens for performance of all the tests of 4.5. The block shall be foamed under restricted, pressure-rise conditions to achieve a density of 22 plus or minus 2 pcf and shall be post heat-cured in the mold between plus 93 and plus 121 degrees C (plus 200 and plus 250 degrees F). A separate test specimen, of sufficient size, shall be fabricated under unrestricted free-rise foam conditions to determine free-rise density in 4.5.3. Not less than five test specimens shall be used for each test.

4.5 Test methods.

4.5.1 Visual examination. All containers and their contents from each lot shall be visually examined for compliance with 3.7 and Section 5.

4.5.2 Test specimens. Preparation of test specimens for quality conformance testing shall be as specified in 4.4 and shall be of the size and dimensions required by the specified test methods.

4.5.3 Density. The free-rise and pressure-rise density shall be determined in accordance with ASTM D 1622 and shall be as specified in Table I.

4.5.4 Compressive strength. The compressive strength shall be determined on a restricted, pressure-rise foamed sample in accordance with ASTM D 1621, Procedure A. The compressive-strength value shall be as specified in Table I.

4.5.5 Tensile strength. The shear strength shall be determined on a restricted, pressure-rise foamed sample in accordance with ASTM D 1623. The tensile-strength value shall be as specified in Table I.

4.5.6 Shear strength. The shear strength shall be determined on a restricted, pressure-rise foamed sample in accordance with ASTM C 273. The shear-strength value shall be as specified in Table I.

4.5.7 Water absorption. The absorption shall be determined on a restricted, pressure-rise foamed sample in accordance with ASTM D 2127. The water absorption value shall be as specified in Table I.

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4.5.8 Humidity aging. The humidity aging shall be determined on a test sample as shown in figure 1 fabricated from each lot of foam material in accordance with the manufacturer's instructions and as follows:

a. The mixed foam shall be weighed to achieve a density of plus or minus 2 pcf.

b. The mold shall be preheated to plus 49 plus or minus 6 degrees Celsuis (C) (plus 120 plus or minus 10 degrees Fahrenheit (F).

c. The test sample shall be temperature stabilized between plus 93 and plus 121 degrees C (plus 200 and plus 250 degrees F) in the mold.

d. After temperature stabilization, the test sample shall be postcured in the mold at this temperature for not less than 30 minutes and then the temperature shall be reduced to plus 49 plus or minus 6 degrees C (plus 120 plus or minus 10 degrees F).

e. When test sample has stabilized at the reduced temperature, it shall be removed from the mold.

f. The test sample shall be measured for confromance to the dimensions identified by a letter in figure 1 and the actual values recorded.

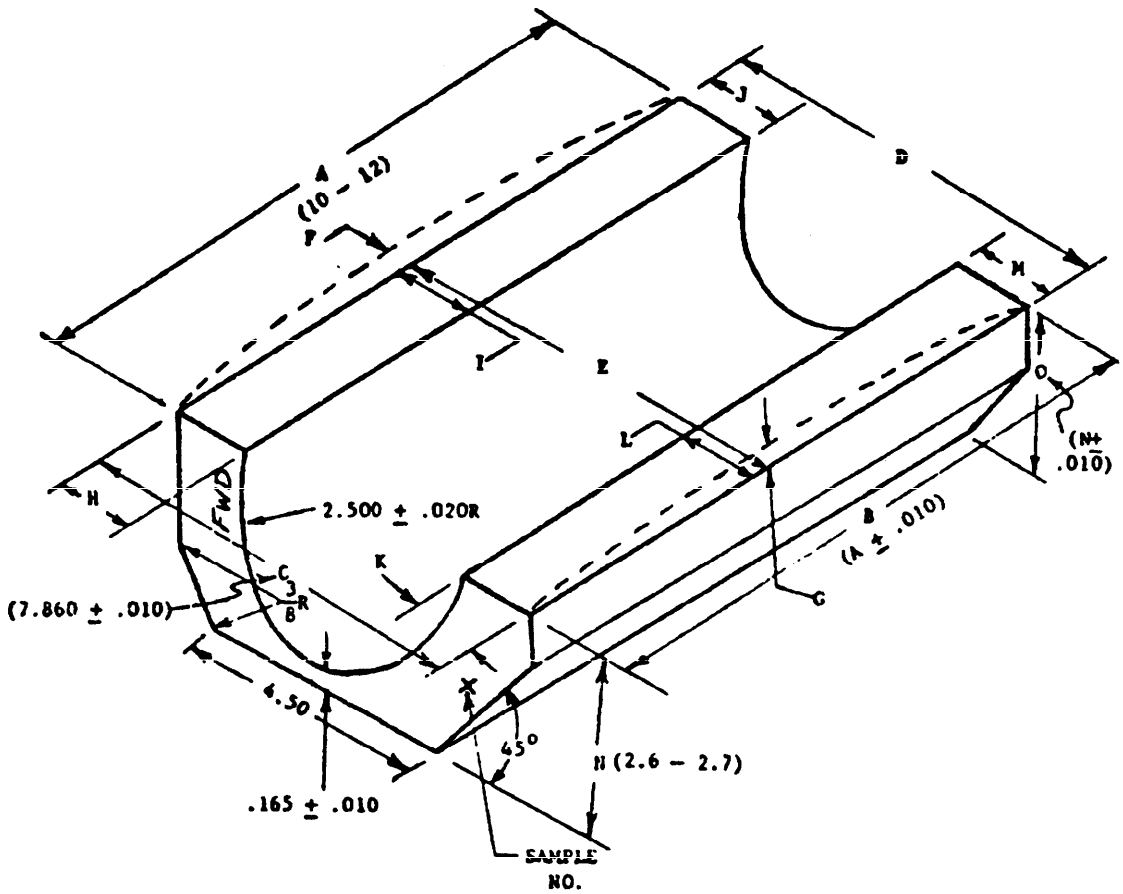
g. The test sample shall then be subjected to a relative humidity between 95 and 100 percent at plus 60 plus or minus 3 degrees C (plus 140 plus or minus 5 degrees F) for not less than 48 hours.

h. Step 4.5.8f shall be repeated and the prehumidity and posthumidity measurements compared. The dimensional changes shall be in accordance with Table II.

Table II

Dimension (see figure 1)	A B	C D E	F G	H I J K L M	N O
Dimensional changes (inch, maximum)	.020	.025	.010	.005	.010

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NOTE: Dimensions shown are for the mold.

Figure 1. Test specimen configuration

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4.5.9 Environmental resistance. The environmental tests shall be performed on a 2-inch cube of restricted, pressure-rise foamed sample post-cured in the mold between plus 93 and plus 121 degrees C (plus 200 and plus 250 degrees F). The percent volume change shall be measured and shall be as specified in Table I. The dimensions of the sample shall be accurately measured before and after exposure to the following environments.

4.5.9.1 Humidity. The sample shall be tested in accordance with MIL-STD-810, Method 507, Procedure I, except that the extreme high test-chamber temperature shall be plus 60 degrees C (plus 140 degrees F) and the controlled relative humidity shall be 85 to 98 percent.

4.5.9.2 High temperature. The sample shall be tested in accordance with MIL-STD-810, Method 501, Procedure I.

4.5.9.3 Low temperature. The sample shall be tested in accordance with MIL-STD-810, Method 502, Procedure I, except that the extreme temperature shall be minus 54 degrees C (minus 65 degrees F) and the soak time shall be 24 hours or until sample temperatures are stablized, whichever is longer.

4.5.9.4 Temperature shock. The sample shall be tested in accordance with MIL-STD-810, Method 503, Procedure I.

4.5.10 Storage life. The contractor shall certify that the materials used in the foam system meet the requirements of 3.6.

4.6 Acceptance criteria.

4.6.1 Preproduction. Failure of any sample to comply with any requirement of this specification shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. The requirements for preservation, packaging, and packing shall be as specified by the procuring activity in the contract or purchase order (see 6.2).

5.2 Marking. Each container shall be marked or tagged in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and location.

b. Material trade name (foam system used).

c. Net weight or volume.

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d. Lot number, batch number, date of manufacture, and expiration date.

e. Number and revision letter of this specification.

f. Recommended storage condition.

6. NOTES

6.1 Intended use. The polyurethane foam covered by this specification is intended for use as a structural component in missile case construction. Foam systems covered are intended to be used in producing the polyurethane foam specified herein.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required and, if so, pertinent details (see 3.1).

c. Request for test data (see 3.2).

d. Minimum lot size, if applicable (see 4.3.2).

e. Requirements for preservation, packaging, packing, and marking (see 5.1).

f. Place of delivery.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-16282A, dated 4 November 1968.

Custodian:
ARMY-MI

Preparing Activity
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Project No. 9330-A603

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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